

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET A — FOUNDATION & CORE CONCEPTS****CHAPTER 03: ATMOSPHERE AND CLIMATE**

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-A-03

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET A - Foundation & Core Concepts):

1. The question paper comprises **25 Multiple Choice Questions** of **1 mark each**.
2. All questions are compulsory. There is no negative marking for incorrect answers.
3. Each question offers four options: **(A), (B), (C), and (D)**. Select the single best answer.
4. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

Q.1 What is the most abundant gas in the Earth's atmosphere, constituting approximately 78% of dry air by volume?

- (A) Oxygen (B) Nitrogen
(C) Argon (D) Carbon dioxide

Q.2 Which gas is essential for respiration in aerobic organisms and constitutes about 21% of atmospheric volume?

- (A) Oxygen (B) Nitrogen
(C) Argon (D) Helium

Q.3 What is the approximate concentration of Carbon Dioxide (CO_2) in the Earth's atmosphere by volume?

- (A) 0.04% (B) 0.93%
(C) 1.5% (D) 0.001%

Q.4 Which is the lowest, densest layer of the atmosphere where almost all meteorological weather phenomena (clouds, rain, storms) occur?

- (A) Troposphere (B) Stratosphere
(C) Mesosphere (D) Thermosphere

Q.5 In the troposphere, temperature typically decreases with increasing altitude at a standard environmental lapse rate of roughly:

- (A) 6.5 °C per 1,000 meters (1 km) (B) 1.0 °C per 1,000 meters
(C) 15 °C per 1,000 meters (D) Temperature remains completely constant

Q.6 Which atmospheric layer contains the Ozone Layer (O_3) that absorbs harmful ultraviolet (UV) solar radiation?

- (A) Stratosphere (B) Troposphere
(C) Mesosphere (D) Exosphere

Q.7 Why do commercial jet aircraft prefer to cruise in the lower stratosphere?

- | | |
|--|---|
| (A) Absence of turbulent clouds and violent convective weather | (B) Presence of extremely high oxygen concentration |
| (C) Gravity is completely absent in the stratosphere | (D) It is the warmest region of the atmosphere |

Q.8 In which atmospheric layer do most incoming meteors burn up upon entering Earth's atmosphere due to friction?

- | | |
|-----------------|------------------|
| (A) Mesosphere | (B) Exosphere |
| (C) Troposphere | (D) Stratosphere |

Q.9 The atmospheric layer containing electrically charged ions that reflects radio waves back to Earth for long-distance telecommunication is the:

- | | |
|---------------------------------------|-----------------|
| (A) Ionosphere (part of Thermosphere) | (B) Troposphere |
| (C) Stratosphere | (D) Ozonosphere |

Q.10 Atmospheric pressure is measured using which scientific instrument?

- | | |
|-----------------|-----------------|
| (A) Barometer | (B) Thermometer |
| (C) Seismograph | (D) Hydrometer |

Q.11 The apparent deflection of moving winds and ocean currents caused by Earth's rotation (deflecting winds to the right in the Northern Hemisphere) is called the:

- | | |
|-----------------------------|------------------------|
| (A) Coriolis effect / force | (B) Centripetal effect |
| (C) Doppler effect | (D) Thermal convection |

Q.12 What is the low-pressure belt situated near the equator where northeast and southeast trade winds converge known as?

- | | |
|---|---------------------|
| (A) Inter-Tropical Convergence Zone (ITCZ) / Doldrums | (B) Horse Latitudes |
| (C) Sub-polar Low Pressure Belt | (D) Polar Highs |

Q.13 High-altitude, narrow bands of swift, meandering westerly air currents flowing in the upper troposphere are known as:

- | | |
|-----------------|-----------------|
| (A) Jet streams | (B) Trade winds |
| (C) Monsoons | (D) Loo |

Q.14 The word 'Monsoon' is derived from the Arabic word 'Mausim', which literally translates to:

- | | |
|------------|---------------|
| (A) Season | (B) Rainstorm |
| (C) Ocean | (D) Windmill |

Q.15 What is the primary physical cause driving the Indian Southwest Monsoon?

- | | |
|---|--|
| (A) Differential heating and cooling of the landmass and surrounding oceans | (B) Constant unchanging temperature between land and sea |
| (C) Earth's magnetic field flipping polarity | (D) Sudden cooling of the equator by polar icebergs |

Q.16 The hot, dry, and gusty summer daytime winds blowing across the northern and northwestern plains of India are locally called:

- | | |
|------------------------------|-------------------|
| (A) Loo | (B) Mango showers |
| (C) Norwesters / Kalbaisakhi | (D) Mistral |

Q.17 Which pre-monsoon showers in Kerala and coastal Karnataka help in the early ripening of mangoes?

- | | |
|-------------------|--------------------------|
| (A) Mango Showers | (B) Blossom Showers |
| (C) Kalbaisakhi | (D) Western Disturbances |

Q.18 The periodic development of a warm ocean current along the coast of Peru that disrupts normal atmospheric circulation and often weakens the Indian monsoon is called:

- | | |
|-----------------|----------------------|
| (A) El Niño | (B) La Niña |
| (C) Gulf Stream | (D) Kuroshio Current |

Q.19 Cyclonic storms originating over the Mediterranean Sea that travel eastward and bring essential winter rainfall to northwestern India are called:

- | | |
|-----------------------------------|-----------------------|
| (A) Western Cyclonic Disturbances | (B) Southwest Monsoon |
| (C) Tropical Cyclones | (D) Tornadoes |

Q.20 Which location on Earth receives the highest average annual rainfall in the world?

- | | |
|----------------------------------|------------------------------------|
| (A) Mawsynram (Meghalaya, India) | (B) Cherrapunji (Meghalaya, India) |
| (C) Mumbai (Maharashtra, India) | (D) Jaisalmer (Rajasthan, India) |

Q.21 Why does Tamil Nadu receive the bulk of its rainfall during October and November rather than during July and August?

- | | |
|---|--|
| (A) It lies in the rain-shadow of the Western Ghats during SW monsoon and receives rain from Northeast (Retreating) monsoon winds | (B) Tamil Nadu is located in the Arctic polar circle |
| (C) The Bay of Bengal completely dries up during July | (D) The Himalayas block winds from reaching southern India in winter |

Q.22 What is the natural process where atmospheric gases trap infrared heat radiated by the Earth, keeping the planet warm enough to sustain life?

- | | |
|-----------------------|------------------------|
| (A) Greenhouse effect | (B) Nuclear fusion |
| (C) Photosynthesis | (D) Acid precipitation |

Q.23 Which of the following is considered a potent greenhouse gas emitted from flooded rice paddy fields and ruminant livestock?

- | | |
|-------------------------------|-----------------------------|
| (A) Methane (CH_4) | (B) Oxygen (O_2) |
| (C) Argon (Ar) | (D) Helium (He) |

Q.24 The total amount of greenhouse gases (especially carbon dioxide) generated directly and indirectly by an individual, organization, or activity is called their:

- | | |
|----------------------|----------------------|
| (A) Carbon footprint | (B) Ecological niche |
| (C) Ozone layer | (D) Biomass index |

Q.25 Which international protocol signed in 1987 successfully phased out the production of Chlorofluorocarbons (CFCs) to protect the stratospheric ozone layer?

(A) Montreal Protocol

(B) Kyoto Protocol

(C) Paris Agreement

(D) Ramsar Convention

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET A)

01 (A) (B) (C) (D)

02 (A) (B) (C) (D)

03 (A) (B) (C) (D)

04 (A) (B) (C) (D)

05 (A) (B) (C) (D)

06 (A) (B) (C) (D)

07 (A) (B) (C) (D)

08 (A) (B) (C) (D)

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13 (A) (B) (C) (D)

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15 (A) (B) (C) (D)

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18 (A) (B) (C) (D)

19 (A) (B) (C) (D)

20 (A) (B) (C) (D)

21 (A) (B) (C) (D)

22 (A) (B) (C) (D)

23 (A) (B) (C) (D)

24 (A) (B) (C) (D)

25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET A)

Q. No.	Correct	Curricular Reference & Conceptual Rationale
Q.1	(B)	Nitrogen makes up roughly 78.08% of Earth's atmosphere by volume, followed by oxygen at 20.95%.
Q.2	(A)	Oxygen accounts for roughly 21% of dry atmospheric air and is vital for animal respiration and combustion.
Q.3	(A)	Carbon dioxide makes up approximately 0.04% (around 420 ppm) of the Earth's atmosphere, functioning as a vital greenhouse gas.
Q.4	(A)	The troposphere extends up to about 8 km at the poles and 18 km at the equator; it contains 75% of atmospheric mass and all weather phenomena.
Q.5	(A)	The normal lapse rate in the troposphere is approximately 6.5 °C drop in temperature for every 1,000 meters (1 km) increase in elevation.
Q.6	(A)	The stratosphere (extending from roughly 12 to 50 km) houses the ozone layer, which shields Earth from damaging UV-B and UV-C solar radiation.
Q.7	(A)	The stratosphere has stable, stratified air without the turbulent vertical convection currents and storm clouds of the troposphere.
Q.8	(A)	The mesosphere (50 to 80 km) is the coldest atmospheric layer; meteors entering from space burn up here due to atmospheric friction.
Q.9	(A)	The ionosphere contains ionized particles that reflect shortwave radio transmissions back to Earth, enabling wireless communication.
Q.10	(A)	A barometer (mercury or aneroid) measures the force exerted per unit area by the weight of the air above.
Q.11	(A)	Ferrel's law states that due to the Coriolis force arising from Earth's rotation, winds deflect to their right in the Northern Hemisphere and left in the Southern Hemisphere.
Q.12	(A)	The ITCZ is a broad equatorial trough of low pressure where converging trade winds ascend, creating calm conditions historically known as the doldrums.
Q.13	(A)	Jet streams are fast-flowing narrow air currents in the upper troposphere (speeds from 110 km/h in summer to 184 km/h in winter) steering weather systems.
Q.14	(A)	'Monsoon' originates from the Arabic word 'Mausim', referring to the seasonal reversal in the direction of winds over a year.
Q.15	(A)	Intense summer heating of the Indian subcontinent creates a deep low-pressure system over northern plains, drawing moisture-laden winds from the high-pressure southern Indian Ocean.
Q.16	(A)	'Loo' refers to the scorching, dusty, dry summer winds blowing during May and June across the northern plains of India.
Q.17	(A)	Pre-monsoon thunderstorms in Kerala and Karnataka are known as 'Mango Showers' because they aid the early ripening of mango crops.
Q.18	(A)	El Niño is a warm ocean current replacing the cold Peruvian current periodically, associated with the Southern Oscillation (ENSO), often suppressing Indian monsoon rains.
Q.19	(A)	Western disturbances originate over the Mediterranean and bring critical winter rain to Punjab/Haryana, highly beneficial for Rabi crops (wheat).
Q.20	(A)	Mawsynram in the Khasi Hills of Meghalaya receives the world's highest average annual rainfall (over 11,800 mm) due to relief orographic funnelling.

Q. No.	Correct	Curricular Reference & Conceptual Rationale
Q.21	(A)	During the Southwest monsoon, Tamil Nadu lies parallel to the Bay of Bengal branch and in the rain shadow of the Western Ghats; it gets rainfall when winds retreat across the Bay of Bengal in winter.
Q.22	(A)	The greenhouse effect occurs when atmospheric gases like CO_2 , methane, and water vapour trap outgoing long-wave terrestrial infrared radiation.
Q.23	(A)	Methane is a powerful greenhouse gas with a high global warming potential, produced by anaerobic decomposition in wetlands, rice paddies, and livestock digestion.
Q.24	(A)	A carbon footprint measures the total greenhouse gas emissions caused directly or indirectly by an individual, product, or institution.
Q.25	(A)	The Montreal Protocol (1987) is a landmark environmental treaty that successfully phased out ozone-depleting substances, notably CFCs.